AgentFleet · Agent Documentation

Dispatch Agent

Role: Fleet Assigner · Pipeline Position: Step 1 of 5 · Endpoint: POST /dispatch/allocate

Receives a cargo delivery request and allocates the best available vehicle and driver.

Overview

The Dispatch Agent is the first agent in the AgentFleet pipeline. It queries available drivers and vehicles, filters by cargo weight capacity, picks the nearest vehicle using the Haversine formula, assigns a driver, creates a trip record, and sets both the vehicle and driver status to **Busy**.

Responsibilities

- Query available drivers and vehicles from the database.
- Filter vehicles by cargo weight capacity (capacity_kg ≥ cargo_weight).
- Select the nearest suitable vehicle to the pickup location using Haversine distance.
- Assign the first available driver to the selected vehicle.
- Create a new trip record in the database.
- Set assigned vehicle and driver status to **Busy**.

API Endpoint

Method	Path	Description
POST	/dispatch/allocate	Allocate driver and vehicle for a new delivery

Request Payload

```json
{
  "pickup":        "chennai",
  "destination":   "bangalore",
  "cargo_weight":  2500
}
```

Response Payload

```
{
  "status": "success",
  "data": {
    "trip_id":  "<uuid>",
    "vehicle":  { "id": "<uuid>", "vehicle_number": "KA-01-AA-1234" },
    "driver":   { "id": "<uuid>", "name": "Speedy Gonzales" }
  }
}
```

Internal Logic Flow

```
allocate_dispatch(pickup, destination, cargo_weight)
   ■
   ■■■ 1. get_available_drivers()         → DriverUnavailableException if none
   ■■■ 2. get_available_vehicles()         → VehicleUnavailableException if none
   ■■■ 3. Filter: capacity_kg >= cargo_weight
   ■■■ 4. parse_coordinates(pickup)        → nearest vehicle via haversine_distance()
   ■■■ 5. Assign first available driver
   ■■■ 6. haversine_distance() + estimate_eta() → distance & duration
   ■■■ 7. create_trip() + update_vehicle_status('Busy') + update_driver_status('Busy')
```

Module Files

File	Purpose
agent.py	Agent registry entry — wraps service call
service.py	Business logic: capacity filtering, nearest vehicle, trip creation
repository.py	DB queries: drivers, vehicles, trip insert, status updates
routes.py	FastAPI router — POST /dispatch/allocate
schemas.py	Pydantic request/response models
prompts.py	AI prompt templates
tools.py	LLM tool definitions

Shared Utilities

Module	Usage
shared.geo.coordinates	parse_coordinates() — city name → lat/lon
shared.geo.distance	haversine_distance() — great-circle km
shared.geo.eta	estimate_eta() — travel time at 45 km/h
shared.exceptions	DriverUnavailableException, VehicleUnavailableException

Error Cases

Condition	Exception	HTTP Code
No drivers with Available status	DriverUnavailableException	503
No vehicles with Available status	VehicleUnavailableException	503
No vehicle meets cargo weight requirement	VehicleUnavailableException	503

■ To reset driver/vehicle availability during testing: GET http://localhost:8000/health/reset